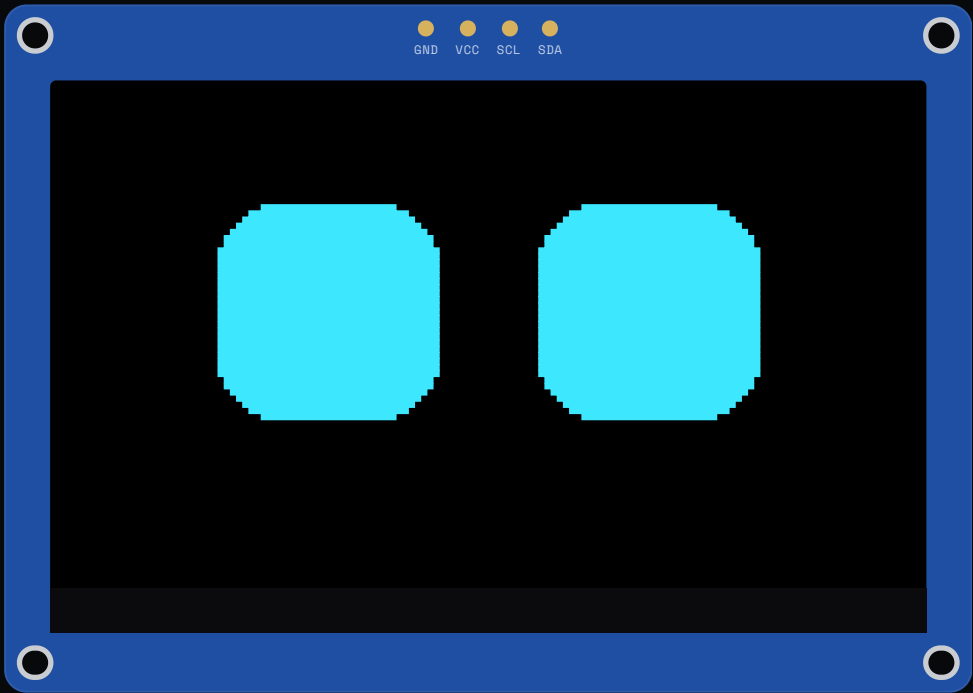


DESKTOP-PET

USER GUIDE AND TECHNICAL NOTES

A small face that lives on the desk. Two rounded eyes on a 0.96 inch OLED glance around, blink, cheer up when you pet it and fall asleep when you ignore it. Plugged into the PC it becomes the face of Aminimal, my voice assistant, and it keeps the time, shows the weather, runs timers and dances to your music.



BOARD

WeAct Black Pill, STM32F401CC or F411CE

DISPLAY

SSD1306 OLED, 128 × 64 pixels, I2C

INPUT

One button, KEY

LINKS

USB, Bluetooth (HC-05), WiFi (ESP-01)

WHAT'S INSIDE

GUIDE

- 01 Meet the pet**
What it is and how it can be connected
- 02 One button**
Tap, double tap, hold
- 03 Left to itself**
Moods, habits and the faces it can pull
- 04 Aminimal's face**
Listening, thinking, speaking
- 05 Music and films**
Headphones, dancing, bars and popcorn
- 06 Screens and cards**
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- 07 Things to try**
Tricks, and the details that are easy to miss

INSIDE

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- 09 Wiring**
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- 10 The firmware**
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- 11 The protocol**
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- 12 How it hears music**
From the speakers to 16 bars and a beat
- 13 WiFi**
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- 14 Building and uploading**
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ABOUT THE PICTURES

Every screen in this handbook is a real capture. The pet was asked for its frame buffer 28 times a second over USB (the `SN` command, chapter 11) and each 1,024-byte frame was drawn back pixel for pixel. Where a figure shows a strip, the frames are in order and the times are measured from the first one. The time, the weather and the timers are example values sent to the pet for the pictures.

MEET THE PET

Desktop-pet is a face for the desk. It runs on a WeAct Black Pill, a thumb-sized board with an STM32 chip on it, and draws on a 128 × 64 OLED about the size of a postage stamp. It has one button, called KEY, and that is the only way to touch it.

On its own it behaves like a small animal: it looks around, blinks, fidgets, enjoys being petted and dozes off when nobody pays attention. Connected to the PC it also becomes the face of Amino, my voice assistant, and a small dashboard for the time, the weather, timers, reminders, and whatever music or film is playing.

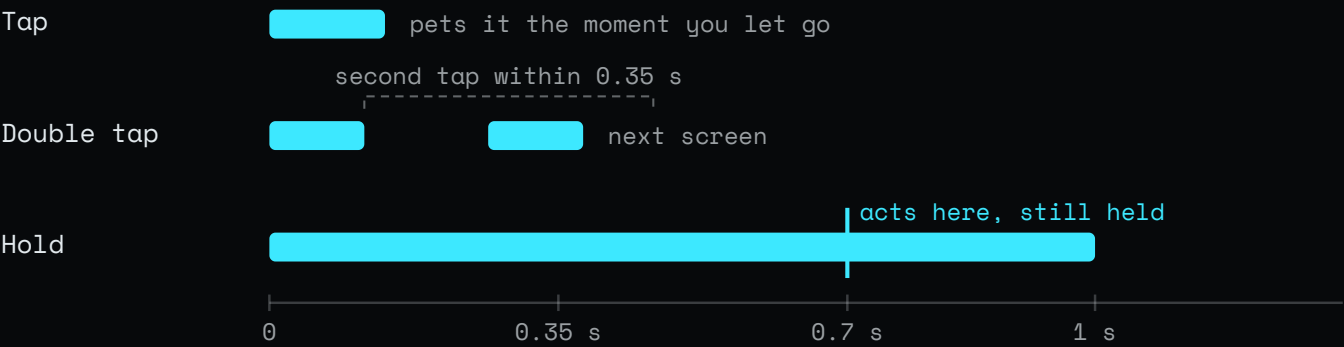
HOW IT'S CONNECTED	WHAT IT DOES
USB cable to the PC	Everything. The fastest link, and the one Amino always prefers.
Bluetooth, on a charger	Everything, a little slower. Amino finds it within 15 seconds of the cable coming out.
A charger, PC off	Still a pet, still keeping the clock it was last given. With the WiFi module working, it fetches its own time and weather.

ONE BUTTON

KEY is the small button beside the USB socket. The other one, NRST, restarts the board. What a press does depends on what is on the screen.

PRESS	ON THE FACE	ON ANY OTHER SCREEN
Tap	Pets it: happy for 2.5 seconds	Next screen
Three taps	Falls in love: two beating hearts. Each tap within 1.5 s of the last.	Three screens on
Double tap	Pets it, then flips to the next screen. Second tap within 0.35 s.	Two screens on
Hold, 0.7 s	Back to the face	On the timer: pause or carry on. Anywhere else: back to the face.

A tap also puts away a card. The screens come round in this order: face, clock, weather, timer (only while one is running) and status. A screen you flip to goes back to the face by itself after 20 seconds.



LEFT TO ITSELF

The pet is never quite still. Every 1.5 to 4 seconds it glances somewhere new, and it blinks every 2.5 to 6 seconds. Every 15 to 40 seconds it also does one small habit, picked at random: a good look left and then right, a little hop, a suspicious squint, or a double blink.



0.00 s



2.03 s



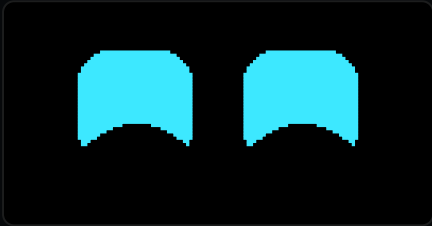
3.22 s



14.00 s

Idle

Glancing right, blinking, looking up to the left, then down. Four moments from 15 seconds of doing nothing.



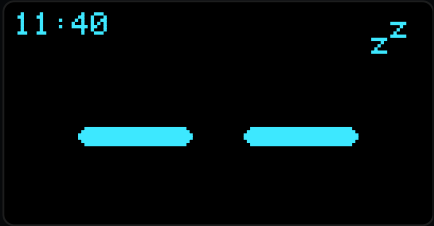
Happy

Tap once. The eyes lift and curve into arcs for 2.5 seconds.



In love

Tap three times, each within 1.5 s of the last. Two hearts beat for 2.5 seconds.

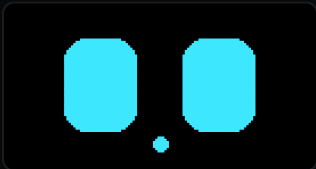


Asleep

Ignore it for 30 seconds, or 10 after 11 pm. The eyes droop to slits, z's float up and the time shows in the corner.

FACES AMINAL CAN ASK FOR

Aminal can make the pet pull any of seven faces for as long as it likes, 3 seconds unless it says otherwise: happy, love, surprised, sad, angry, wink and sleepy. The pet also looks surprised by itself when a timer runs out or a film suddenly gets loud.



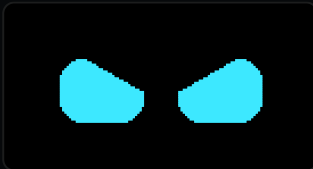
Surprised

Tall eyes and a small round mouth.



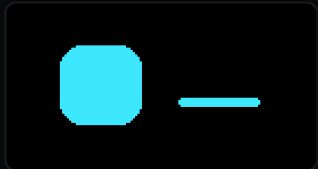
Sad

The outer corners droop.



Angry

The inner corners pull down.

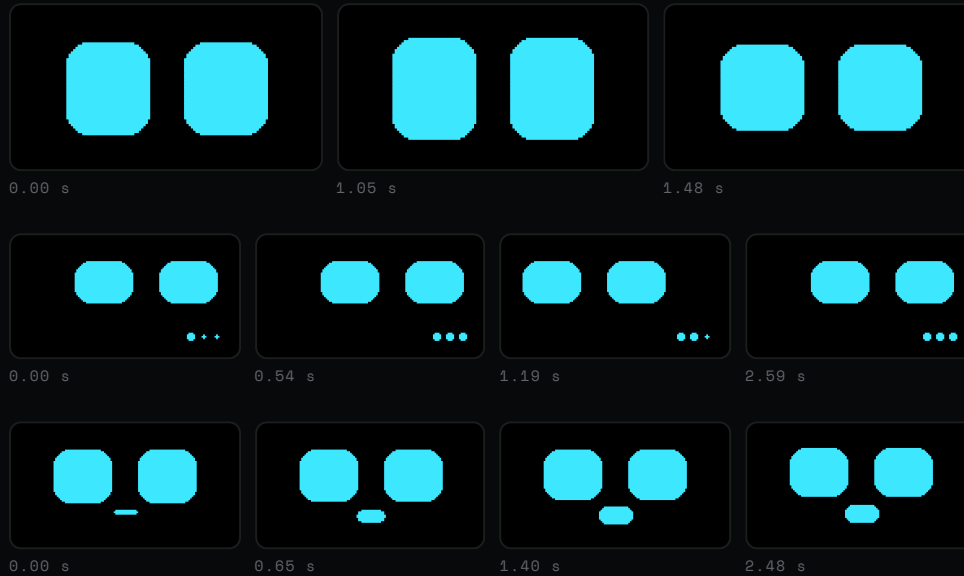


Wink

One eye closes to a line.

AMINAL'S FACE

While Aminoal runs, the pet shows what it is doing. The bridge (chapter 08) sends Aminoal's state and the loudness of the voice up to 20 times a second, so the eyes keep up with speech.



Listening

Eyes wide open, swelling with the loudness of your voice.

Thinking

Looking up and off to one side, switching sides every second or so, while three dots fill in.

Speaking

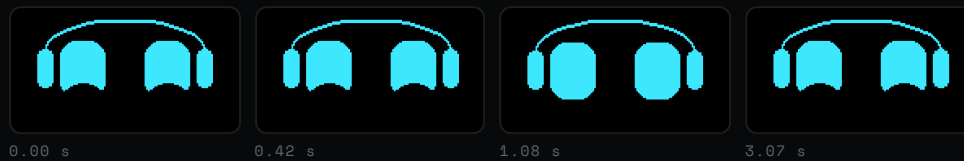
The eyes lift to make room for a mouth that opens with Aminoal's voice.

IT FOLLOWS THE CONVERSATION

- Start talking to Aminoal while the clock or the weather is up, and the pet flips back to the face to listen.
- Ask for the weather or a timer, and it shows that screen while Aminoal answers.
- Answers that carry a reminder, mail or what's next on the calendar arrive as a card (chapter 06).
- When Aminoal connects, the pet greets it with a happy face. When Aminoal closes, the pet waits 8 seconds for another message and then carries on by itself.

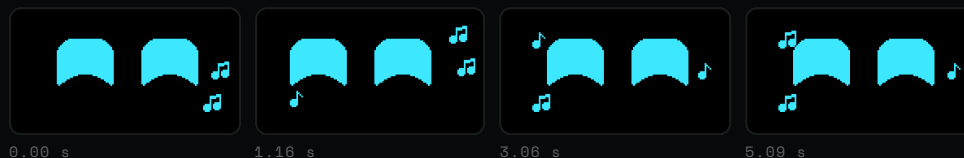
MUSIC AND FILMS

When music plays, the pet joins in with one of three moods, picked at random and changed every 25 to 45 seconds. It rarely picks the same mood twice in a row.



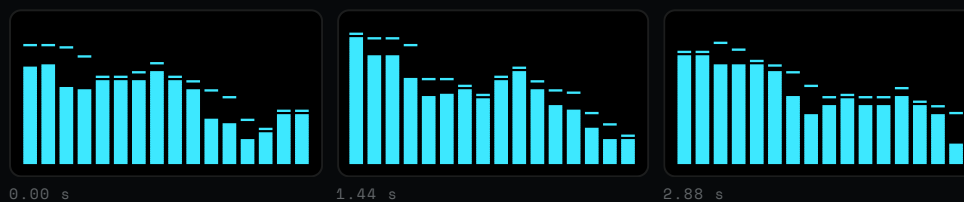
Headphones

Headphones on, eyes shut and happy, nodding on every beat. Every nine seconds or so it opens its eyes for a look around (third frame).



Dancing

Sways from side to side and hops on the beat. A note floats up on every second beat.

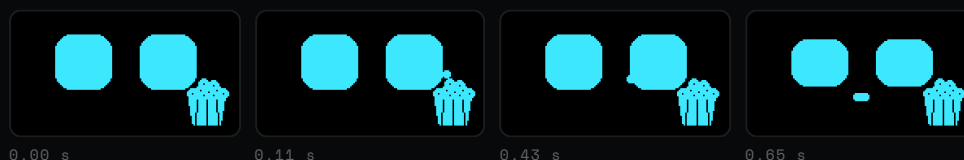


Visualizer

The whole screen turns into 16 bars of what the speakers are playing, low notes on the left, with peak marks that fall back slowly. Captured with test bars sent over USB.

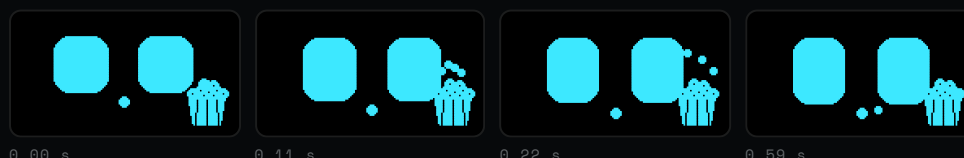
FILMS

Put on Netflix, Prime Video or Hotstar and it looks up at the screen with a striped bucket of popcorn.



Popcorn

Every 3.5 to 7 seconds it flicks a kernel out of the bucket, catches it and chews.



A loud scene

When the sound suddenly jumps, so does the pet: surprised eyes, and four kernels fly out of the bucket.

HOW IT TELLS THEM APART

Aminal decides from the app and the window title, never from the sound itself.

Spotify is music. A browser tab that says Netflix is a film. A video player showing an .mp3 is music. Anything it can't name for certain, such as a game, a video call or a YouTube video, gets no reaction at all.

- A new kind of sound has to play for 2 seconds before the pet reacts, so a notification chime is not a song.
- Silence has to last 6 seconds before it stops, so the gap between two tracks doesn't end the dance.
- When several apps make sound, the loudest one it can name wins, so a chime during a film doesn't take the popcorn away.
- Without beats from the PC, it dances to a steady 115 beats a minute of its own.

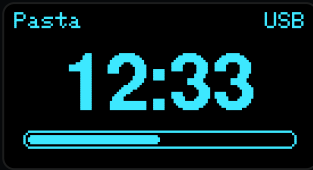
SCREENS AND CARDS



Clock
The time in big digits, the date above it, and a bar that fills across each minute.



Weather
An animated icon, the temperature, today's high and low, and the place. After three hours without news it adds "(old)".



Timer
Counts down by itself between updates, over a bar that drains. Paused, it blinks like an oven.



Status
The link, what Aminal is doing, WiFi, how fresh the time and weather are, and how long it has been up.

WEATHER ICONS

Every icon moves: the sun's rays turn, rain and snow fall, fog drifts and lightning flickers. From 7 pm to 6 am a clear sky shows a moon with twinkling stars. The numbers are the WMO weather codes Open-Meteo reports.



Clear
Code 0



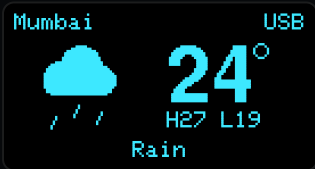
Partly cloudy
Codes 1 and 2



Overcast
Code 3



Fog
Codes 45 and 48



Rain and showers
Codes 61 to 67 and 80 to 82; drizzle (51 to 57) falls lighter



Snow
Codes 71 to 77, 85 and 86



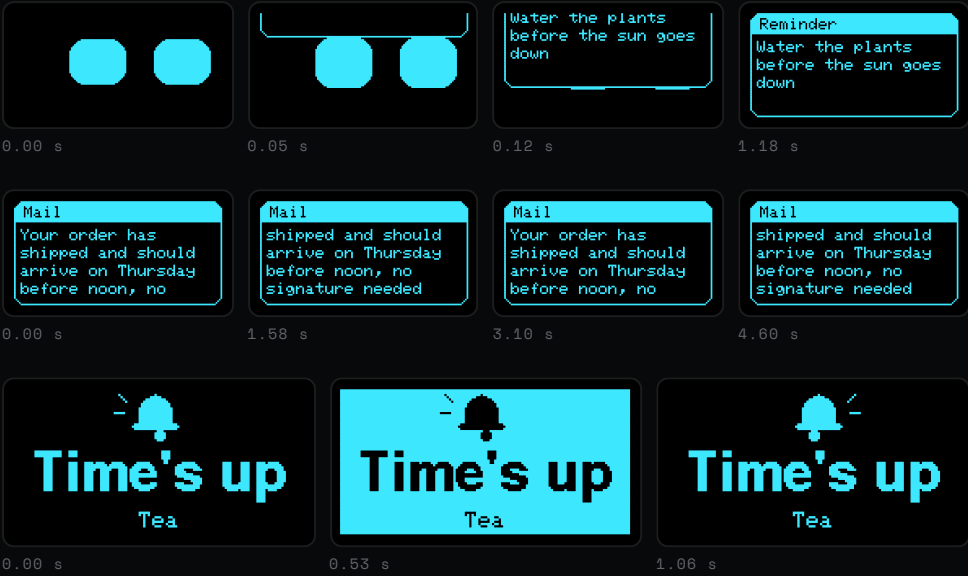
Thunderstorm
Codes 95 to 99



Clear, at night
Code 0 from 7 pm to 6 am

CARDS

Reminders, mail and what's next on the calendar slide down over whatever is on screen and stay for a few seconds. A tap puts one away early.



A card

Slides down from the top in 150 milliseconds.

A long card

Four lines fit. Longer text scrolls up a line every 1.5 seconds.

Time's up

When a timer runs out, the whole screen rings: a swinging bell, the timer's name, and a flash every half second until you tap it. The flash is the display inverting itself, which a frame-buffer capture can't see, so it is added back here.

THINGS TO TRY

Some of what the pet does only shows up when you go looking for it.

WITH YOUR FINGER

Pet it three times, fast.

Taps less than 1.5 seconds apart add up, and the third turns its eyes into hearts.

Leave it alone.

After 30 seconds it droops off to sleep. One tap wakes it.

Check on it late at night.

After 11 pm it falls asleep after only 10 seconds and shows the time while it sleeps, so it works as a night clock.

Double tap.

It walks through the clock, the weather, the timer and the status screen, then comes back to the face by itself.

Hold it on the timer.

The timer pauses and blinks. Hold again to carry on.

WITH AMINAL

Talk while the clock is up.

It flips back to the face to listen.

Speak up.

The eyes grow with your voice.

Close Aminimal.

Eight seconds later the pet is on its own, clock still running.

Pull the cable out.

If the pet is on a charger, Aminimal finds it over Bluetooth within 15 seconds. Plug the cable back in and it moves back to USB by itself.

WITH SOUND

Play a long playlist.

The mood changes every 25 to 45 seconds: headphones, dancing, or the visualizer.

Watch the headphones.

Every nine seconds or so it peeks out.

Skip a track.

A gap shorter than 6 seconds doesn't stop the dance.

Get a message during a film.

The chime doesn't take the popcorn away.

Join a video call, or play a game.

Nothing happens, on purpose.

SMALL DETAILS

The bar under the clock

is the seconds, filling across the minute.

The sun's rays

turn, slowly.

Old weather

says so: after three hours without an update the screen adds "(old)".

A ringing timer

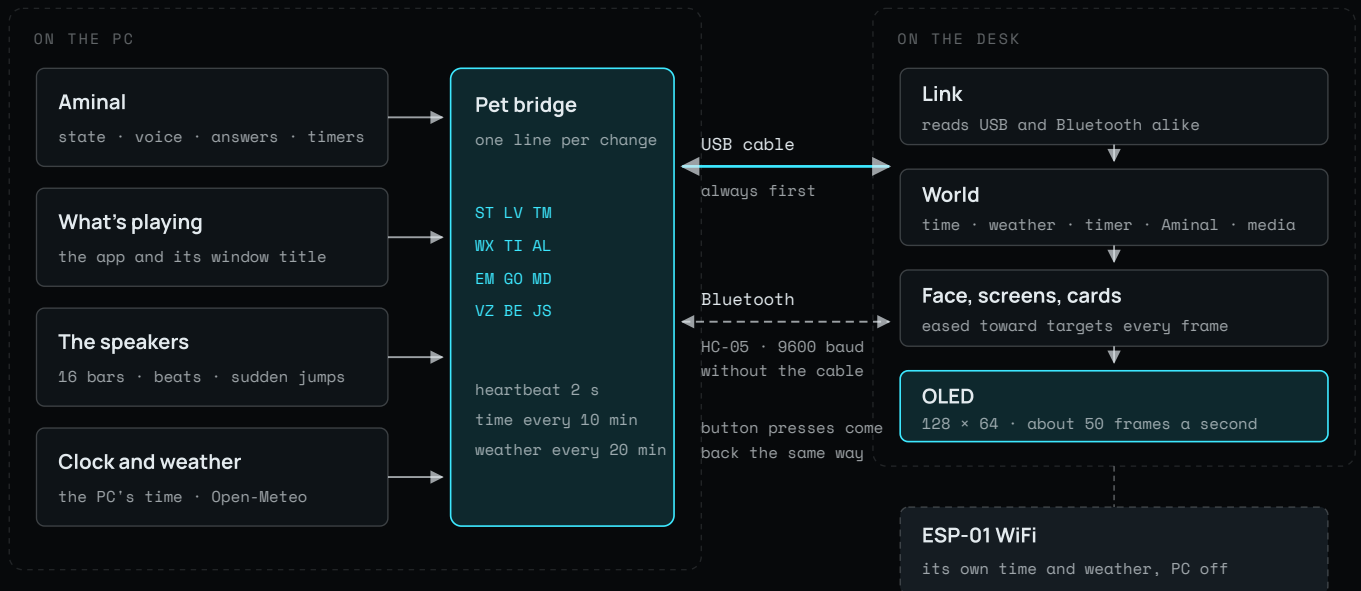
outranks everything: a reminder can't cover it.

Before anyone tells it the time,

the clock says "waiting for time" instead of guessing.

HOW IT FITS TOGETHER

Aminal never talks to the pet itself. A small bridge program runs beside it, reads what Aminal is doing 25 times a second, and turns every change into one short line of text. The pet reads those lines, keeps what they say in one shared state it calls the World, and draws that state 50 times a second.

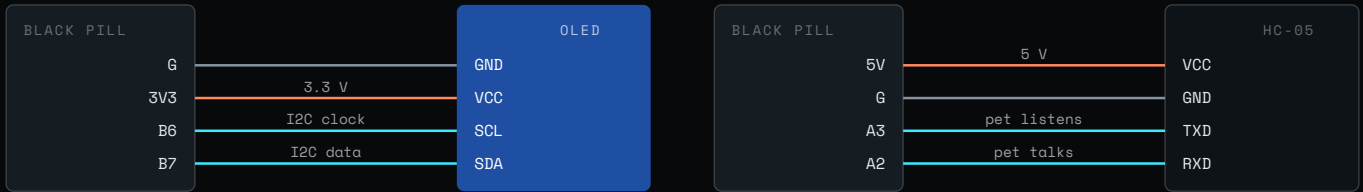


FINDING THE PET

- The bridge only talks to a port that answers **HI** with **PET**, so it never sends lines to the wrong device.
- USB always wins. While the cable isn't there it tries the Bluetooth port every 15 seconds; opening the port of a pet that is switched off keeps Windows busy for seconds, so it doesn't try more often.
- It sends a heartbeat every 2 seconds. After 8 seconds of silence the pet decides it is on its own.
- It follows Aminal's process. When Aminal exits, the bridge says goodbye to the pet and exits too.
- Nothing is sent twice: each kind of line goes out only when its value changes, apart from the time, which is refreshed every 10 minutes.

WIRING

PART	WHAT IT'S FOR
WeAct Black Pill	The brain: an STM32F401CC or F411CE, USB-C, the KEY and NRST buttons, and a blue LED on PC13.
0.96" SSD1306 OLED	The face: 128 × 64 pixels over I2C, at address 0x3C.
HC-05	Optional. The Bluetooth link to AminaL.
ESP-01	Optional. Its own time and weather over WiFi.
AMS1117-3.3, 470 µF	Power for the ESP-01, from the 5 V pin.

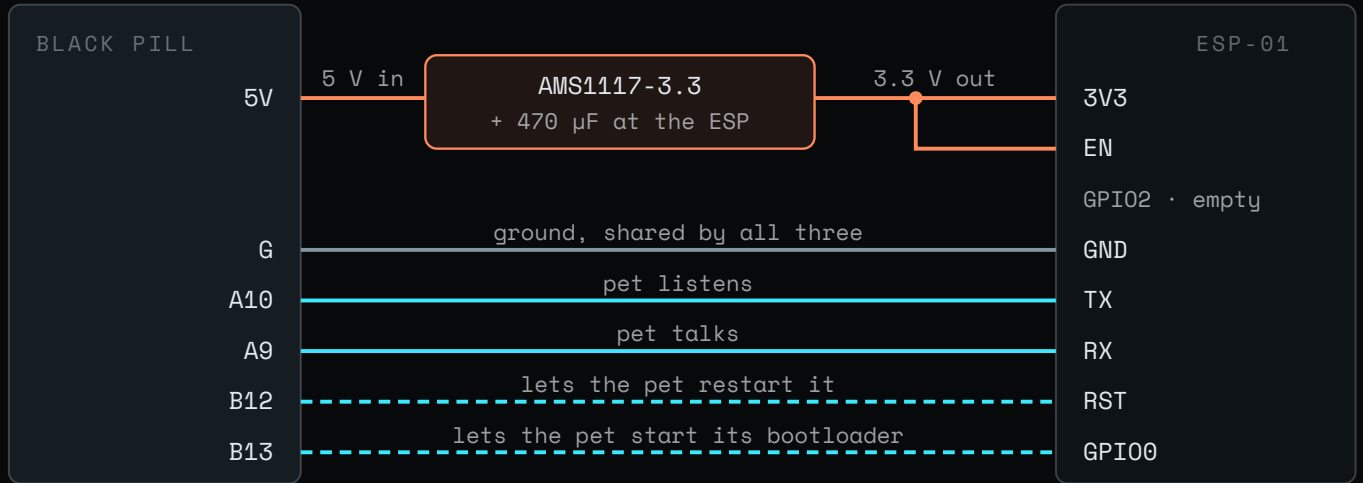


The OLED

Powered from the board's own 3.3 V. It draws little, so that is fine.

The HC-05

Six pins in a row; STATE and EN stay empty. Its red LED blinks fast when it is ready to pair. Pair it with the PC (PIN 1234 or 0000), not a phone.



The ESP-01

Eight pins in two rows of four. With the chips facing you and the antenna up, one row holds GND, GPIO2, GPIO0 and RX, the other TX, EN, RST and 3V3. GND and 3V3 sit in opposite corners.

POWER THE ESP-01 ON ITS OWN

The ESP-01 is a 3.3 V part, and 5 V damages it. The Black Pill's 3.3 V pin can't feed it either: the radio pulls bursts of about 300 mA, and on this unit that made the module garble its replies and restart halfway through a network scan, even at lowered transmit power. An AMS1117-3.3 regulator fed from the 5 V pin, with a 470 µF capacitor across the module's 3.3 V and ground, gives it the current it needs.

THE FIRMWARE

C++ on the Arduino framework (STM32duino), built with PlatformIO. Adafruit's SSD1306 and GFX libraries draw the screen, and FreeSans Bold sets the big digits.

ONE PASS OF THE LOOP

- 01 Read every line waiting on USB and Bluetooth, up to 512 bytes a port per pass, so a flood of lines can't starve the animation.
- 02 Move the WiFi module's state machine on by one step. It never waits for the module.
- 03 If 20 ms have passed since the last frame: read the button, check the link and the timer, update the face, draw the screen and any card, and send the 1,024-byte frame to the OLED over I2C.

That comes to about 50 frames a second, as fast as the I2C bus carries them.

EASING

No expression is drawn directly. Every frame, the eyes' position, width and height move part of the way toward a target:

```
current += (target - current) * 0.25 // position, width, and most heights
height += (target - height) * 0.6   // blinking, or closing by more than 10 px
height += (target - height) * 0.45   // while listening, to keep up with the voice
```

So glances glide, blinks snap shut, and falling asleep is a slow droop. Every face in this handbook is just a different set of targets.

WHO GETS THE FACE

The first of these that applies decides what the eyes do. Music and films step aside whenever Aminimal is listening, thinking or speaking.

1 A face Aminimal asked for	2 Happy, after a tap	3 Listening
4 Thinking	5 Speaking	6 Music
7 A film	8 Asleep	9 Idle: glances, blinks, habits

TIMINGS

WHAT	WHEN	WHAT	WHEN
Glance	every 1.5–4 s	Music mood	25–45 s
Blink	every 2.5–6 s, 130 ms	Its own tempo	115 bpm, 520 ms
Habit	every 15–40 s	Popcorn	every 3.5–7 s
Happy after a tap	2.5 s	Back to the face	after 20 s
Hearts	3 taps under 1.5 s apart	Link lost	8 s of silence
Falls asleep	30 s; 10 s, 11 pm–7 am	Hold / double tap	0.7 s / 0.35 s

Every timer is compared with `reached(now, t)` rather than `now > t`, so the pet keeps blinking when the millisecond counter wraps round after 49.7 days.

THE PROTOCOL

One short line of text per message, the same on USB and Bluetooth, and simple enough to type into a serial monitor. A line longer than 159 characters is dropped whole rather than read in pieces.

LINE	MEANING
HI	Who are you? The pet answers PET desktop-pet 6.
PG	Heartbeat, answered P0.
ST idle listening thinking speaking off	What Aminimal is doing
LV 0-100	How loud the voice is, for the eyes and the mouth
TM <unix> <offset s>	The time, and the UTC offset
WX <temp> <code> <hi> <lo> <place>	The weather, with its WMO code
TI <left> <total> <paused> <label>	The timer, or TI - for none
AL <secs> <title> <text>	A card, or AL - to take it down
EM <face> [ms]	Pull a face: happy, love, surprised, sad, angry, wink, sleepy
G0 <screen> [secs]	Show face, clock, weather, timer or status
MD music [style] watch -	What's playing. A music style (headphones, dance, bars) can be asked for by name.
VZ <16 hex digits>	Spectrum bars, 0 to f each, low notes first
BE	A beat
JS	A sudden loud moment in a film
SN	Answered SN <hex>: exactly what is on the OLED
L0 <lat> <lon> <place>	Where the weather is for, kept in flash for the WiFi
WF <ssid><tab><password>	USB only: join a WiFi network
ES talk flash [baud], ES reset	USB only: the cable straight through to the ESP-01, for debugging

The pet sends events back up whichever link spoke last: EV pet, EV next <screen>.

EV hold, EV hold timer, EV dismiss, EV wifi ok <ip> and EV wifi fail.

A FEW SECONDS OF CONVERSATION

```

→ HI
← PET desktop-pet 6
→ TM 1790658725 19800           // 10:42 on 29 Sep, India's +5:30
→ WX 29 2 32 24 Mumbai
→ ST listening
→ LV 64
→ ST speaking
→ MD music dance
→ BE
← EV pet                       // someone tapped it

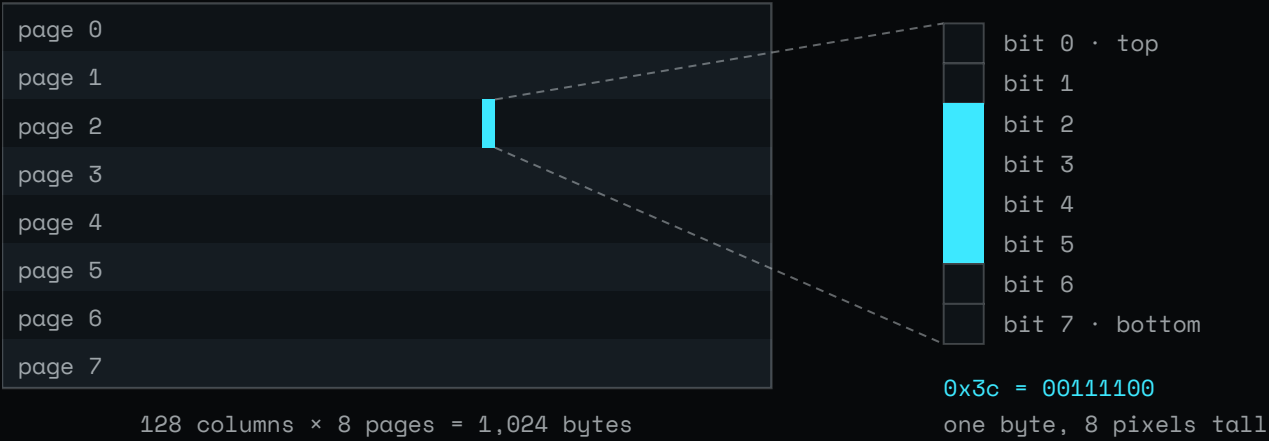
```

SNAPSHOTS: HOW THESE PICTURES WERE MADE

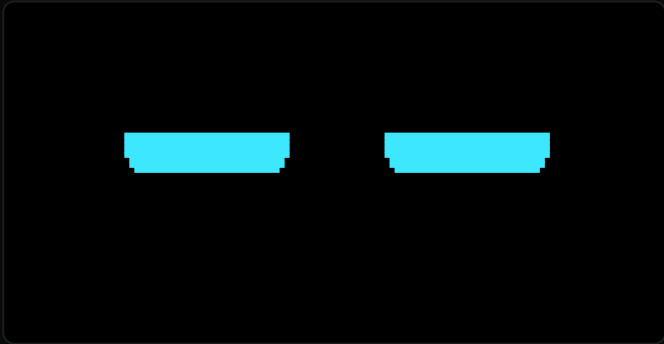
SN answers with the whole frame buffer as 2,048 hex digits. The buffer is 8 pages of 128 bytes, and each byte is a column of 8 pixels with the top pixel in the lowest bit. Here is the start of a real reply, taken while the pet was in love, and a piece of page 3, where the hearts are:

```
SN 0000000000000000000000000000000000000000000000000000000000000000... // page 0: the top 8 rows, dark
... page 3, from column 24:
ffffffffffffffffffffffffffffffffffffffffffffffffffff7f1f0000000000000000
... 2,048 digits in all
```

Over USB one snapshot takes about 36 ms, so the pet can be filmed at 28 frames a second while it keeps drawing its own 50. Over Bluetooth, at 9600 baud, one takes about 2 seconds. Each strip in this handbook is a run of those frames, decoded and drawn back pixel for pixel.



Decoded
The reply above, drawn back: all 8 pages, 128 × 64 pixels.



Page 3 alone
Only bytes 384 to 511, the 8 rows through the middle of the hearts.

HOW IT HEARS MUSIC

All of this runs on the PC, inside the bridge. The pet only ever receives the results as MD, VZ, BE and JS lines.

STEP	WHAT HAPPENS
1 · What's playing	Once a second, Windows' per-app volume meters list the apps making sound (a peak above 2%) . Each is named by its app, then its window title, then the file it plays. The loudest one that gets a name wins.
2 · Listening in	Only while music or a film is on, the bridge records what the speakers play through WASAPI loopback (PyAudioWPatch).
3 · Bars	Each block of 1,024 samples, about 23 ms, goes through an FFT into 16 bands spaced by pitch from 50 Hz to 12 kHz. The top end is lifted 4.5 dB per octave so the treble moves as much as the bass, and the bars show the loudest 40 dB under an automatic gain. Sent as VZ 20 times a second, 10 over Bluetooth.
4 · Beats	Energy between 40 and 150 Hz rising to 1.45 times its average over the last second, at least 0.28 s after the last beat. Sent as BE.
5 · Jumps	A block 4 times louder than the median of the last 5 seconds, at most once every 8 seconds. Sent as JS, for films only.

THE SOUND STAYS PUT

Each block becomes 16 numbers and two yes-or-no flags and is thrown away. Nothing is recorded, saved or sent anywhere, and the listener only runs while the pet has music or a film to show.

NAMED FOR CERTAIN, OR NOT AT ALL

CLUE	MUSIC	FILM
The app	Spotify, Apple Music, Tidal, Deezer, foobar2000, MusicBee...	Netflix, Prime Video, Disney+, Hotstar, JioCinema apps
A browser's title	Spotify, YouTube Music, SoundCloud, JioSaavn, Gaana, Wynn	Netflix, Prime Video, Hotstar, JioCinema, Disney+, SonyLIV, ZEE5, Apple TV
A video player's file	.mp3, .flac, .m4a, .wav, .ogg, .opus, .aac, .wma	anything else it plays

Browsers are recognised even when they aren't on the list, by the audio helper process every Chromium and Firefox browser uses. That is how Comet was picked up.

WIFI

The ESP-01 keeps the factory AT firmware it shipped with, and the pet drives it with plain text commands. Once it has joined a network it makes one plain HTTP request to Open-Meteo every 20 minutes. That single reply carries the time (in its Date header), the UTC offset and today's weather, so a pet on a charger keeps its clock and forecast with the PC switched off.

```
start → ATE0 → AT+CWMODE_DEF? → AT+CIPMUX=0 → AT+CIFSR → idle
idle → AT+CIPSTART="TCP","api.open-meteo.com",80 → AT+CIPSEND → GET → +IPD ... CLOSED
// every 20 min; again after 5 min if it failed; connection checked every minute
```

JOINING

Close Aminimal and run the bridge's WiFi setup. It offers the network the PC is on, asks for the password without showing it, and sends both down the USB cable only: the pet refuses them over Bluetooth. The module remembers the network and rejoins by itself. It only sees 2.4 GHz networks.

WHAT IT KEEPS

Where the weather is for (sent by Aminimal as L0) and the UTC offset are saved in the last sector of the Black Pill's flash, so they survive a power cut. The WiFi password isn't: only the module holds it.

STATUS ON 29 SEP 2026

The code is finished and the module answers, but on this unit its radio browns out on the Black Pill's 3.3 V (chapter 09). WiFi starts working once the module has its own regulator.

Talking to Aminimal over WiFi as well needs the ESP-01 reflashed with a small relay program. Reflashing it through the pet failed, because its ROM loader restarts on every sync, so that waits for a USB-serial adapter.

BUILDING AND UPLOADING

WeAct boards ship with the WeAct HID bootloader in their first 16 KB. Windows sees it as a plain HID device (0483:572A), so it needs no driver, and it avoids the chip's own USB bootloader, which often fails with Device Descriptor Request Failed.

- 01 **Install PlatformIO.** In VS Code, open Extensions, search for PlatformIO IDE and install it.
- 02 **Get the uploader.** Download `WeAct_HID_Flash-CLI.exe` from WeAct's repository and save it as `tools/WeAct_HID_Flash-CLI.exe`. It is WeAct's own closed tool, so it isn't in the repository.
- 03 **Open the project.** File, Open Folder, `Desktop-pet`. PlatformIO fetches the STM32 toolchain and the display libraries itself.
- 04 **Enter the bootloader.** Hold KEY, tap NRST, wait for the blue LED to blink, then let go of KEY. A brand new board is already in it.
- 05 **Upload.** Click the arrow in the blue status bar.
- 06 **Wake it.** The eyes appear. If they don't, tap NRST.

The default build targets the F401CC and also runs on an F411CE. For the F411's full 100 MHz, check the square chip and switch `default_envs` to `blackpill_f411ce`.

WHERE EVERYTHING LIVES IN FLASH



That is the F401CC's 256 KB, drawn to scale. The app is linked to start right after the bootloader, at 0x08004000, and has 16 KB of room before it would meet the saved settings.

WHEN SOMETHING'S OFF

WHAT YOU SEE	LIKELY CAUSE
Blue LED blinking fast, screen blank	The OLED isn't answering on I2C. Check its four wires, then try address 0x3D in <code>src/main.cpp</code> .
Upload can't find the board	It isn't in the bootloader. Hold KEY, tap NRST, let go when the LED blinks.
No HID device at all, even holding KEY	The bootloader was erased by a DFU or ST-Link upload to 0x08000000. Reflash it from WeAct's repository, or build without the flash offset.
Garbage on the screen	The module may be an SH1106, common on 1.3 inch boards, which needs a different library.
Eyes never react to Amino	The status screen says Link none: the bridge isn't running, or a serial monitor has the port.
No light on the HC-05	Almost always a loose VCC or GND wire.
ESP-01 answers AT, then garbles and restarts	A brownout. Give it its own AMS1117-3.3 and a 470 µF capacitor (chapter 09).

A MAP OF THE CODE

FILE	WHAT'S IN IT
<code>src/pet.h</code>	The shared World: time, weather, timer, Amino's state, media, the current card
<code>src/main.cpp</code>	Setup, the frame loop, the button, switching screens
<code>src/face.cpp</code>	Eyes, moods, habits, and the listening, thinking, speaking, music and film faces
<code>src/media.cpp</code>	Headphones, music notes, the visualizer and the popcorn
<code>src/screens.cpp</code>	Clock, weather icons, timer, status and the cards
<code>src/link.cpp</code>	The protocol, read from USB and Bluetooth alike, and the ESP-01 passthrough
<code>src/wifi.cpp</code>	The ESP-01 driven through its AT firmware: joining, the request, reading the reply

DOCUMENT HISTORY

REVISION	DATE	WHAT CHANGED
Rev. 1	29 Sep 2026	First edition, for firmware v6: the user guide, the technical notes, and real captures of every face and screen.